Harry Graves

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Dual British & Irish National - DBS Level 2 Security Clearance

Education & Awards:

University of Oxford, St Anne's College **October 2021 – July 2025**

MEng Engineering Science: Upper second-class honours (2:1)

- Master's thesis (78%): egocentric semantic scene understanding with Meta's *Project Aria* AR glasses; trained an LSTM to detect hand-object interactions from hand- and eye-tracking data
- Third-year project (76%): music source separation using transformers and panning angle isolation
- Specialising in machine learning, computer vision & signal processing; proficient in Python and C++
- Institute of Engineering and Technology (IET) Future Talent Awards scholar; Fung scholar

Bishop Vesey's Sixth Form, Sutton Coldfield **September 2019 – July 2021**

A-levels: Mathematics A*, Physics A*, Chemistry A*

- National runner-up in RAF engineering award *Schools Aerospace Challenge*; awarded £1000 prize

Bishop Walsh Catholic School, Sutton Coldfield **September 2014 – July 2019**

GCSEs: eight grade 9s including English and Mathematics, one grade 8, one grade 6, FSMQ Mathematics

Research and Work Experience:

Co-founder & ML Engineer, Autotone **June 2025 – Present**

- Commercialisation of third-year design project, machine learned guitar effect parameter estimation
- Development of novel hybrid waveform-spectrogram transformer models for guitar tone embedding
- Winners of *Reuben College Entrepreneurship Forum 2025* pitch competition; awarded £30000 prize

Machine Vision Engineering Intern, Quell **August 2025 – October 2025**

- Research and development of novel machine learning techniques for anti-air systems
- Used a simulator to generate synthetic data; experimented with its use in pretraining and finetuning
- Flew several quadcopter and fixed-wing drones; captured and annotated images of them for training

Research Intern, Oxford Robotics Institute **September 2024 – October 2024**

- Developed ROS2 (Robot OS) compatibility for Meta's *Project Aria* AR research glasses, for use with an online SLAM (Simultaneous Localisation and Mapping) system
- Benchmarked localisation performance against offline bundle-adjustment methods

Research Intern, Huazhong University of Science and Technology **June 2024 – August 2024**

- Developed and tested several neural rendering technologies including 3D Gaussian Splatting (3DGS) and NeRFs, to model the University's campus in Wuhan, China
- Prototyped software to use these photorealistic 3D reconstructions for virtual campus tours

Engineering Design Intern, IPS – Integrated Project Services **June 2023 – September 2023**

- Developed design skills including CAD, engineering drawing and technical specification writing
- Worked on and visited several active projects including new vaccine and drug production facilities

Skills & Interests:

Sports: St Anne's College football club president – responsibilities included looking after the club's budget and deciding where it should be spent, attending meetings, running weekly training and organising matches

Music: Vocalist and guitarist for local band Head Height since 2020, responsible for liaising with promoters and other local bands to organise several gigs in and around Birmingham

Languages: English - Native; Spanish – B2 Intermediate